MD MERAJ ANSARI

B.Tech Student | AI/ML Engineer | Innovator

+91-8639961862 | mdmeraj3500@gmail.com | Hyderabad, Telangana | LinkedIn

PROFILE SUMMARY

Dedicated and innovative second-year B.Tech student at Symbiosis Institute of Technology with strong expertise in AI/ML, computer vision, and full-stack app development. Multiple hackathon winner with published IEEE research. Proven leader through roles in IEEE student branch, event coordination, and blockchain club. Passionate about building technology solutions that create real-world impact.

EDUCATION

Bachelor of Technology (B.Tech) *2024 – Present*

Symbiosis Institute of Technology, Hyderabad | Symbiosis International Deemed University

TECHNICAL SKILLS

Languages: Python (90%), C (85%), R (80%), Java, HTML & CSS, Dart/Flutter
Libraries & Frameworks: NumPy, Pandas, Matplotlib, Seaborn, OpenCV, FastAPI, YOLOv8
Tools & Platforms: Google Cloud, IoT Development, Blockchain (Algorand), Firebase, Convex, OSRM, Flutter
Domains: Machine Learning, Computer Vision, Deep Learning, NLP, IoT, Full-Stack App Development

EXPERIENCE & LEADERSHIP

Chair, IEEE Industry Applications Society (IAS) *2024 – Present*

Symbiosis Institute of Technology, Hyderabad Student Branch

- Lead the student branch chapter, organizing technical events, workshops, and seminars on industry applications
- Promoted student membership and engagement in IEEE activities; fostered innovation in electrical and electronics engineering
- Attended IEEE ICASP 2025 – International Conference on Acoustics, Speech and Signal Processing

Co-Lead, Algorand Blockchain Club *2024 – Present*

Symbiosis Institute of Technology, Hyderabad

- Co-led the student blockchain club, fostering peer-to-peer learning and technical skill development
- Conducted workshops on Algorand Blockchain technology, smart contracts, and automated bot creation

Event Coordinator, ANVESHAN 2025 *March 2025*

Symbiosis Institute of Technology, Hyderabad – Annual Technical Fest

- Coordinated a diverse range of events including treasure hunts, neon parties, esports competitions, and DJ nights
- Led a team through strategic planning, stakeholder management, and multi-event execution across two days (8–9 March 2025)

HACKATHON ACHIEVEMENTS

ECE Department Hackathon – 1st Place | KL University, Azizanagar Campus | Oct 8, 2025

- Won ₹15,000 prize in a 24-hour hardware + software innovation challenge focused on hardware-software integration
- Demonstrated problem-solving under pressure and strong team collaboration and leadership skills

IEEE Hackathon Winner – IEEE Leadership Summit 2025

- Secured victory in the IEEE Hackathon; 3rd major hackathon title with core team: Raj Aryan, Pramit Panigrahi, Smaran Jaianand
- Consistently demonstrated collaborative spirit, technical talent, and innovation across multiple competitions

TechSynapse Hackathon Winner | Team Neo | 24-Hour Sprint

- Built 'ArthNiti' – an AI-powered financial advisor that transforms raw financial data into smart, actionable insights

- Delivered a working product under intense 24-hour competition conditions with strong team collaboration

RESEARCH & PUBLICATIONS

IEEE Paper: Sign Language Recognition Using AI and Machine Learning

2025 IEEE 3rd Global Conference on Wireless Computing and Networking (GCWCN) | IEEE Bombay Section

- Presented published research aimed at making communication more accessible for the hearing-impaired through AI/ML
- Conference Chairs: Dr. Dnyaneshwar S. Mantri & Dr. Sachin D. Babar | 22–23 November 2025

IEEE Paper: Plant Disease Detection Using YOLOv8m

Symbiosis Institute of Technology, Hyderabad | Under Review – IEEE Conference

- Trained YOLOv8m model on ~142K images across 38 disease classes achieving mAP50 = 99.0%, Precision = 98.7%, Recall = 98.4%
- Built FastAPI backend + Firebase frontend with ElevenLabs TTS voice output for real-world farmer use on a Tesla T4 GPU

FEATURED PROJECTS

SwiftSky – Weather-Aware Delivery Routing App

Flutter | OSRM | OpenMeteo | Convex | MapTiler | Socket.io | OneSignal

- Built a delivery routing app for Indian riders (Zomato/Swiggy style) with a Comfort Score combining weather (40%), distance (40%), and traffic (20%)
- Integrates real-time weather data to recommend the safest and fastest delivery route with live OTP verification (Fast2SMS)
- Full-stack: Flutter frontend, Convex cloud backend, Socket.io real-time updates, and OneSignal push notifications

Plant Disease Detection System – YOLOv8m

Python | YOLOv8 | FastAPI | Firebase | ElevenLabs TTS | Google Colab

- Real-time plant leaf disease detection across 38 classes with 99.0% mAP50; trained on Tesla T4 GPU (~142K images)
- Complete pipeline: HSV leaf-presence gating, entropy filtering, texture analysis, and voice output via ElevenLabs TTS

eCarbonKarma – Eco-Reward Mobile App

Flutter | Algorand Blockchain | Firebase | ML | Computer Vision

- Gamified sustainability app: users earn Algorand tokens by photographing waste disposal and logging fitness activities
- Blockchain-backed reward system incentivizes eco-friendly behavior with transparent, tamper-proof token distribution
- Built during a 20-hour Open Innovation Hackathon; framed as a 'Climate Consequence Engine' with ML pattern detection

Sign Language Detection System

Python | YOLOv8 | OpenCV | Computer Vision

- Real-time sign language recognition with end-to-end pipeline: data collection, labeling, training, validation, and deployment
- Published research paper at IEEE GCWCN 2025 based on this system

Obstacle-Avoiding Robot Car

Arduino | C++ | Robotics | IoT

- Autonomous vehicle using ultrasonic sensors for intelligent real-time navigation and obstacle avoidance

ArthNiti – AI Financial Advisor

Python | NLP | Machine Learning | FastAPI

- Hackathon-winning AI financial advisor that converts raw financial data into actionable investment insights

CERTIFICATIONS & PARTICIPATION

- Machine Learning Certification – Codesoft

- Python Coding Certification – InternPE
- Google Cloud Skill Boost Badges: Arcade Base Camp (April & May 2025), Arcade Trivia (April 2025, Week 1 & 2), Arcade Certification Zone (April 2025)
- Certificate of Participation – GUVI IIT Bombay AI Impact Summit 2026 (Mission Upskill India Pre-Summit, organized by HCL GUVI & India AI)
- IEEE ICASP 2025 – International Conference on Acoustics, Speech and Signal Processing
- IEEE GCWCN 2025 – Published paper presenter, IEEE Bombay Section